

Tran Duc Son

AI & SOFTWARE ENGINEER

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SUMMARY OF QUALIFICATIONS

- AI Department Manager at Bkav with 8+ years spanning security and AI. Strong in malware analysis, reverse engineering, and building AI/ML systems: NLP, LLM, semantic search, ASR, Text-to-Speech, and generative models; experienced with big data pipelines, vector databases, and cloud deployment. Leads teams and drives strategy for AI products from research to production. Comfortable across backend services (FastAPI), containers (Docker, K8s), and mobile/web when needed.

SKILLS

Languages:

Python, C++, JavaScript, SQL

AI / Machine Learning:

NLP, LLM, Generative AI, Semantic Search, Embedding, ASR, Text-to-Speech, Malware detection models, Computer Vision

Security & Low-level:

Malware behavior analysis, Reverse engineering, Disassembly, Detection/removal development

Data & Back-end:

Big Data, Data pipelines, REST API, Cloud AI, FastAPI, Flask. **SQL:** MySQL, MariaDB, PostgreSQL. **NoSQL:** MongoDB. **Vector DB:** Milvus, FAISS, Elasticsearch / OpenSearch.

DevOps: Docker, Kubernetes (K8s)

Mobile & Web:

Next.js, React, Flutter, Android, iOS, Full-stack web, Automation flows, No-code/Low-code, AI Agent Platform

Management & Other:

Team leadership, Project management, AI product development, R&D

PROFESSIONAL EXPERIENCE

2021 – 2026

Bkav

Manager → AI Department Manager (Data Science)

Technology Stack:

LLM, Generative AI, ASR, Text-to-Speech, AI Agent Platform, No-code/Low-code, Python, NLP, Cloud AI, REST API, Team Leadership

Main responsibilities:

- Led and grew the Data Science department; continued developing and shipping Core AI products (malware detection, data pipelines, spam detection)
- Researched and implemented Large Language Models and Generative AI from scratch; deployed ASR and Text-to-Speech models on internal systems
- Drove AI Agent Platform initiatives (no-code/low-code) for building agents and automated workflows
- Responsible for development strategy, research, and application of AI across Bkav products

2020 – 2021

Bkav

Intern — Data Science Department

Technology Stack:

Malware research, Big Data, AI Core models, Python, Data pipelines, NLP

Main responsibilities:

- Researched malware and big data; built, researched, optimized, and deployed AI Core models for malware detection
- Developed data collection systems on Facebook, Instagram, online newspapers, and other platforms
- Contributed to the company's AI-based spam email detection system

2018 – 2020

Bkav

Intern — Malware Analysis

Technology Stack:

Malware behavior analysis, Disassembly, Reverse engineering, C/C++

Main responsibilities:

- Analyzed malware behavior and wrote detection/removal functions
- Performed disassembly and reverse engineering of malicious binaries

COMPANY PROJECTS

Vietnamese Legal Management System and Legal Support Chatbot (2025 – Present)

Role: Team Lead / Architect

Description: Built a system to manage Vietnamese legal documents and integrated an NLP-based chatbot for legal lookup and basic legal advice. The system helps users search, compare documents, and get quick answers to common legal questions.

Team size: 6

My responsibilities:

- Analyze and design system architecture
- Deploy NLP pipeline and integrate chatbot

- Review and optimize models, deploy to production

Techstack: AI, NLP, Web, Python, FastAPI, MongoDB, Vector Database, REST API

Vietnamese Wikipedia Data Collection and Semantic Search System (2024 – 2025)

Role: ML Engineer / Research

Description: Collected, cleaned, and normalized data from Vietnamese Wikipedia; built embedding and semantic search pipeline for meaning-based information retrieval.

Team size: 6

My responsibilities:

- Collect and preprocess Wiki data
- Build text embedding and semantic search models
- Integrate into internal search product

Techstack: AI, NLP, Data, Python, FastAPI, MongoDB, Vector Database, Elasticsearch

Research, Build, Improve and Optimize AI Core Model for Bkav Pro Security AI Cloud

(2020 – Present)

Role: Research Engineer / Team Lead

Description: Research and develop AI Core model for malware detection, behavior analysis, and device protection in Bkav Pro Security. Deploy and optimize pipeline on AI Cloud.

Team size: 6

My responsibilities:

- Analyze and design AI model for malware detection
- Continuously research, improve, and optimize models
- Deploy inference on Cloud, ensure performance and updates for new threats

Techstack: AI, Security, Cloud, Python, C++, FastAPI, REST API

AI Agent Platform — Low-code / No-code platform for automation and AI systems

(2025 – Present)

Role: Team Lead / Architect

Description: Low-code/no-code platform for deploying automated workflows and AI-powered systems. Used to deliver: automated content publishing for vnreview.vn; AI customer care system; and the “Virtual Legal Assistant” (Trợ lý ảo pháp chế) chatbot for the Ministry of Science and Technology.

Team size: 6

My responsibilities:

- Design and evolve the AI Agent Platform (low-code/no-code)
- Deploy and maintain automated posting system for vnreview.vn
- Build and integrate AI customer care and chatbot systems
- Deliver Virtual Legal Assistant chatbot for Ministry of Science and Technology

Techstack: AI, LLM, NLP, No-code/Low-code, FastAPI, MongoDB, Web, Automation

PERSONAL PROJECTS

Driving Theory — Test generation and driving theory exam simulator (2025)

Role: Full-stack Developer

Description: App for driving theory practice and mock exams: random test generation, exam simulation aligned with real tests, QR Code support. Available on Android and iOS.

Team size: 1

Links: [Google Play](#) · [App Store \(iOS\)](#)



Android



iOS

My responsibilities:

- Analyze and design system
- Develop Android and iOS apps and QR Code scanning
- Implement test generation and grading logic

Techstack: Android, iOS, FastAPI, MongoDB

BKSafe — Phone tracking and malware scanning tool (2024 – 2025)

Role: Mobile Developer

Description: Android app for device status tracking, phone location/tracking, and integrated malware scanning. QR Code support for connecting and managing devices.

Team size: 1

Links: [Google Play \(Android\)](#)



Android

My responsibilities:

- Develop Android application
- Integrate tracking, location, and malware scanning features
- Implement QR Code device connection

Techstack: Android, FastAPI, MongoDB

ChefEasy — Cooking guide website & app (2025)

Role: Full-stack Developer

Description: Website and mobile app with recipes and cooking guides, search by ingredients and difficulty level. AI cooking assistant, timer, save & share recipes.

Team size: 1

Links: [Google Play \(Android\)](#) · [App Store \(iOS\)](#) · [Website](#)



Android

iOS

Website

My responsibilities:

- Design and develop web interface
- Build recipe search system

Techstack: Web, FastAPI, MongoDB

Vietnam and Global Law tracking system + Legal support Chatbot (2025)

Role: Full-stack Developer

Description: Web system aggregating and tracking Vietnamese and international legal documents, with a chatbot for Q&A and basic legal lookup.

Team size: 1

My responsibilities:

- Collect and manage legal data
- Build NLP chatbot for legal lookup

Techstack: Web, AI, FastAPI, MongoDB, Milvus, OpenSearch, Semantic Search, LLM, OpenRouter API, OpenAI API, Vertex AI API

EDUCATION

Hanoi University of Science and Technology (2016 – 2021)

Major: Information Security (An toàn thông tin)

TOEIC: 600

INTERESTS

AI technology, Coding, Reading, Gaming

REFERENCES

[GitHub](#) | [LinkedIn](#)